



# University of Asia Pacific

## Department of CSE

### Business and Entrepreneurship Project Report

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## **Navigating the Future of Urban Farming: “A Smart Aquaponics Approach”**

**Date:** 18 May 2026

**To:** Towfiq Ahmed  
Lecturer  
Department of Business Administration  
University of Asia Pacific

**Subject:** Submission of Business Plan for *FarmX – Navigating the Future of Urban Farming*

**Dear Sir,**

With due respect, we would like to submit our business plan for your kind review and evaluation. We are a team of Computer Science and Engineering students developing an innovative startup named **FarmX**, focused on transforming urban agriculture through smart rooftop farming technology.

### **About Our Business**

FarmX is a sustainable urban farming solution that utilizes rooftop spaces to produce fresh, chemical-free green vegetables and **fishes (aquaponics-based system)** using smart farming technology, IoT-based monitoring, and efficient resource management. Along with food production, we also provide rooftop farming setup services.

### **Market Opportunity**

The demand for safe, organic, and fresh vegetables and fish is rapidly increasing in urban areas like Dhaka. Traditional supply chains often fail to ensure quality, freshness, and affordability. FarmX aims to bridge this gap by producing food locally within the city using unused rooftop spaces.

### **Total Startup Capital**

- Estimated Initial Investment: **≈ 6,00,000 BDT**
  - Rooftop farming setup (~4,00,000 BDT)
  - Supporting infrastructure, seeds, labor, and logistics (~2,00,000 BDT)

### **Projections**

- Expected break-even point: **2<sup>nd</sup> Year**
- Revenue generated from:
  - Total Annual Revenue per Unit → **₳4,00,000**
  - Annual Operational Cost (after setup) → **₳80,000**
  - Annual Operating Profit per Unit → **₳3,20,000**

We sincerely appreciate your time and consideration in reviewing our business plan. Your feedback and guidance will be highly valuable for improving and refining our venture.

**Thank you for your consideration.**

**Sincerely,**  
Uap\_RedHawks

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# 1. Executive Summary

## Business Concept

FarmEx is a Dhaka-based smart urban farming startup focused on transforming unused rooftop spaces into sustainable food production units through aquaponics and IoT-enabled agriculture. Our integrated farming system combines fish cultivation and vegetable production in a single water-efficient ecosystem, allowing both food sources to grow together naturally while minimizing water waste and eliminating the need for chemical fertilizers.

FarmEx aims to solve major urban challenges including food safety concerns, lack of fresh organic produce, and inefficient rooftop utilization in densely populated cities like Dhaka. The system uses smart sensors and IoT technology to monitor water level, temperature, humidity, pH balance, and oxygen supply in real time, ensuring stable production and higher farming efficiency.

### Our target customers include:

- Health-conscious urban consumers seeking fresh, chemical-free vegetables and fish
- Rooftop owners who want to generate income from unused rooftop spaces
- Restaurants and local grocery businesses looking for reliable organic food suppliers
- Institutions and communities interested in sustainable urban agriculture solutions

FarmEx generates revenue through direct food sales, rooftop farming setup services, maintenance services, and B2B partnerships with restaurants and retailers.

## Market Opportunity

Bangladesh is experiencing rapid urbanization and increasing demand for safe, organic, and traceable food. Consumers in Dhaka are becoming more health conscious and are willing to pay premium prices for fresh and chemical-free products.

Traditional supply chains often fail to ensure food freshness, safety, and transparency. FarmEx addresses this gap by producing food directly within urban areas and delivering products from rooftop to customer within a very short time.

### Key Market:

- Rising health awareness among urban consumers
- Increasing demand for organic and pesticide-free food
- Growth of smart agriculture and sustainable farming technologies
- Large amount of unused rooftop spaces in urban Bangladesh
- Expansion opportunities in restaurants, supermarkets, and online grocery platforms

FarmEx positions itself as a sustainable and scalable urban farming solution capable of supporting future smart-city food systems.

## Competitive Advantage

| Feature                         | FarmEx Advantage                                                    |
|---------------------------------|---------------------------------------------------------------------|
| <b>Smart Aquaponics System</b>  | Simultaneous fish and vegetable production using a single ecosystem |
| <b>IoT Monitoring</b>           | Real-time monitoring of pH, temperature, oxygen, and water quality  |
| <b>Chemical-Free Production</b> | No harmful fertilizers                                              |
| <b>Rooftop Utilization</b>      | Converts unused urban rooftops into productive farming spaces       |
| <b>Dual Revenue Model</b>       | Revenue from both food production and farming setup services        |
| <b>Faster Supply Chain</b>      | Fresh rooftop-to-door delivery within 45 min                        |
| <b>Sustainable Farming</b>      | Lower water usage and environmentally friendly production           |

## Business Model – Revenue Streams

| Revenue Stream                | Description                             | Target Customer     |
|-------------------------------|-----------------------------------------|---------------------|
| Fresh Vegetables & Fish Sales | Direct sale of chemical-free products   | Urban households    |
| Rooftop Setup Service         | Smart aquaponics installation service   | Building owners     |
| IoT Maintenance Service       | Sensor monitoring and maintenance       | Setup clients       |
| Restaurant Supply             | Regular fresh food supply contracts     | Restaurants         |
| Training & Consultation       | Urban farming workshops and consultancy | Institutions & NGOs |

## Financial Highlights

Fish Production Revenue 600 kg yearly ≈ ₹3,60,000

Vegetable Production Revenue ≈ ₹40,000

Total Annual Revenue per Unit ₹4,00,000

| Metric                 | Year 1                            | Year 2                    | Year 3                   |
|------------------------|-----------------------------------|---------------------------|--------------------------|
| Total Revenue (BDT)    | 4,00,000                          | 4,00,000                  | 4,00,000                 |
| Operational Cost (BDT) | 6,00,000                          | 80,000                    | 80,000                   |
| Net Profit / (Loss)    | (-2,00,000)                       | 1,20,000                  | 3,20,000                 |
| Break-Even Status      | Initial Investment Phase          | Partial Recovery Achieved | Fully Profitable         |
| Monthly Income         | —                                 | ≈ 30,000                  | ≈ 30,000+                |
| Revenue Sources        | Fish + Vegetables + Setup Service | Recurring Farm Revenue    | Stable Recurring Revenue |

## Funding Requirement

FarmEx seeks an initial seed investment of **BDT 4,00,000** to support:

- Rooftop farm infrastructure setup
- IoT sensor and smart monitoring system
- Fish tanks, irrigation, and aquaponics equipment
- Initial operational and marketing expenses

The investment will help FarmEx launch a scalable smart urban farming ecosystem capable of generating sustainable long-term revenue while improving urban food security.



## 2. Industry Analysis

### Industry Size, Growth Rate, and Sales Projections

The global urban farming market was valued at approximately **USD 210 to 280 billion in 2026** and is projected to grow significantly over the next decade due to rising food safety concerns, urbanization, and demand for sustainable agriculture solutions.

In Bangladesh, the demand for fresh, chemical-free, and traceable food is rapidly increasing, especially in urban areas like Dhaka. Consumers are becoming more health-conscious and are willing to pay premium prices for safe and organic food products. At the same time, thousands of rooftops in Dhaka remain unused, creating a major opportunity for rooftop farming businesses like FarmEx.

#### FarmEx projects stable recurring revenue through:

- Fish and vegetable sales
- Rooftop farming setup services
- Restaurant and B2B supply partnerships

#### Sales Projection (Per Rooftop Unit)

| Year   | Revenue (BDT) | Cost (BDT) | Net Profit / Loss |
|--------|---------------|------------|-------------------|
| Year 1 | 4,00,000      | 6,00,000   | (-2,00,000)       |
| Year 2 | 4,00,000      | 80,000     | 1,20,000          |
| Year 3 | 4,00,000      | 80,000     | 3,20,000          |

FarmEx expects to achieve break-even within approximately **2 years**.

### Industry Structure

The urban food and fresh produce market in Bangladesh is currently divided into several major segments:

| Segment                                | Characteristics                             | FarmEx Position                          |
|----------------------------------------|---------------------------------------------|------------------------------------------|
| Traditional Markets (Kacha Bazar)      | Low traceability, pesticide concerns        | Alternative solution                     |
| Supermarkets (Shwapno, Agora, Unimart) | Premium but dependent on external suppliers | Direct competitor                        |
| Online Grocery Platforms (Chaldal)     | Delivery-focused model                      | Direct competitor                        |
| Organic Food Stores                    | Limited supply and scalability              | Alternative                              |
| Rooftop Farming Startups               | Small-scale and fragmented                  | Competitive advantage through technology |

**FarmEx differentiates itself by combining:**

- Rooftop farming
- Aquaponics
- IoT monitoring
- Direct delivery
- Chemical-free production

into one integrated urban farming ecosystem.

**Nature of Participants****The industry currently includes:**

- Traditional vegetable suppliers
- Grocery chains and supermarkets
- Online grocery delivery companies
- Small rooftop gardening businesses
- Agri-tech startups

Most competitors focus only on selling food products, while FarmEx operates as both:

1. A food producer and supplier
2. A smart rooftop farming technology provider

**Key Success Factors**

The success of FarmEx depends on several important factors:

- Consistent supply of fresh and chemical-free food
- Reliable IoT monitoring and smart farming technology
- Strong customer trust and product transparency
- Efficient rooftop partnership management
- Fast rooftop-to-door delivery system
- Cost-efficient farm setup and maintenance
- Strong social media and brand awareness

**Industry Trends**

Several industry trends support the growth of FarmEx:

| <b>Trend</b>                | <b>Impact on FarmEx</b>                |
|-----------------------------|----------------------------------------|
| Rising health awareness     | Higher demand for organic food         |
| Food safety concerns        | Increased trust in local farming       |
| Smart agriculture adoption  | Supports IoT-based farming             |
| Urban rooftop utilization   | Large expansion opportunity            |
| Sustainable living movement | Increased demand for eco-friendly food |
| Online delivery culture     | Easier customer acquisition            |

## **Long-Term Prospects**

FarmEx plans to expand from Dhaka into other major cities such as Chittagong and Sylhet like that flood effected cities through scalable rooftop farming units.

### **Future goals:**

- Expanding rooftop farm partnerships
- Building a city-wide smart farming network
- Introducing AI-based yield optimization
- Developing a full mobile monitoring application
- Creating franchise and licensing opportunities
- Supplying restaurants, supermarkets, and institutions at scale

In the long term, FarmEx aims to become a leading smart urban farming and sustainable food technology company in Bangladesh.

## 3. Company Description

### Company History

FarmEx started as a simple idea during our **1st and 2nd semester university life when our team discussed** the growing problems of unsafe food, urban pollution, and unused rooftop spaces in Dhaka city. At that time, the concept was only at the brainstorming level, and we did not perform detailed research or business analysis.

Later, **during our 3.1 semester Business Project course**, we decided to seriously develop and execute the idea as a complete startup concept. After conducting market analysis, industry research, financial planning, and competitor analysis, we realized that urban rooftop farming combined with smart technology could become a sustainable and scalable business opportunity in Bangladesh.

As a result, FarmEx was developed as a smart urban farming startup that transforms unused rooftop spaces into productive aquaponics farms using IoT-based monitoring systems. The project combines fish cultivation and vegetable farming within a single eco-friendly system to produce fresh, chemical-free food for urban consumers.

FarmEx aims to solve several urban challenges simultaneously:

- Food safety concerns
- Lack of organic produce
- Underutilized rooftop spaces
- Sustainable food production
- Smart agriculture adoption in Bangladesh

The company is designed not only as a food business but also as a technology-driven sustainable urban farming ecosystem.

### Mission Statement

#### Mission

To provide safe, chemical-free, and sustainable food to urban households by converting unused rooftop spaces into smart farming ecosystems using modern technology.

#### Vision

To become Bangladesh's leading smart urban farming and aquaponics platform by integrating sustainability, technology, and modern agriculture into everyday urban life.

#### Tagline

"সবুজ যখন ঘরের অতিথি।"

## Products and Services

FarmEx offers both food products and smart farming services.

### Core Products

| Product          | Description                            | Target Customer        |
|------------------|----------------------------------------|------------------------|
| Fresh Vegetables | Rooftop-grown chemical-free vegetables | Urban households       |
| Fresh Fish       | Aquaponics-based fish production       | Families & restaurants |

### Services

| Service                 | Description                               | Target Customer            |
|-------------------------|-------------------------------------------|----------------------------|
| Rooftop Farm Setup      | Complete smart aquaponics installation    | Building owners            |
| IoT Monitoring System   | Smart sensor and mobile monitoring system | Setup clients              |
| Farm Maintenance        | Regular system maintenance and support    | Rooftop partners           |
| Training & Consultation | Urban farming and aquaponics workshops    | NGOs, schools, individuals |

## Current Status

FarmEx is currently in the development and execution preparation stage. The team has already completed:

- Business model development
- Market research and competitor analysis
- Financial projection planning
- Smart farming system design
- Initial rooftop farming concept validation

Currently, working on:

- Pilot rooftop implementation
- IoT monitoring system development
- Partnership discussions with rooftop owners
- Funding and operational preparation

Although the business has not officially launched yet, FarmEx is actively preparing for pilot execution and future commercial operations.

## Legal Status and Ownership

FarmEx plans to register as a **Private Limited Company** under the Registrar of Joint Stock Companies and Firms (RJSC), Bangladesh.

The company will operate according to:

- Bangladesh Food Safety Authority (BFSA) guidelines
- Sustainable urban farming standards
- Smart agriculture compliance requirements

Ownership of the company will initially remain with the founding team members, with future opportunities for seed investors and strategic partners.

## Key Partnerships

| Partner Type                 | Role                                      | Status                       |
|------------------------------|-------------------------------------------|------------------------------|
| Rooftop Owners               | Provide rooftop spaces for farming        | Ongoing discussion           |
| Organic Seed Suppliers       | Supply farming materials and seeds        | Planned                      |
| Restaurants & Cafés          | B2B supply partnership for fresh products | Future target                |
| Technology Partners          | IoT sensor and monitoring support         | Under development            |
| Delivery & Logistics Support | Product delivery operations               | Planned future collaboration |

## 4. Market Analysis

### Market Segmentation and Target Market Selection

FarmEx targets both B2C and B2B customer groups in urban areas, especially within Dhaka city.

#### B2C Segments

| Segment                      | Profile                                                  | Priority  |
|------------------------------|----------------------------------------------------------|-----------|
| Health-Conscious Families    | Families seeking safe and chemical-free food             | Primary   |
| Organic Food Consumers       | Customers willing to pay more for fresh organic products | Primary   |
| Fitness & Wellness Community | Gym-goers and health-focused individuals                 | Secondary |
| Rooftop Owners               | Building owners with unused rooftop spaces               | Secondary |

#### B2B Segments

| Segment                 | Profile                                             | Priority  |
|-------------------------|-----------------------------------------------------|-----------|
| Restaurants & Cafés     | Businesses needing fresh vegetables and fish supply | Primary   |
| Grocery & Retail Stores | Premium food retailers and supermarkets             | Secondary |
| Corporate Offices       | Companies promoting healthy and sustainable food    | Tertiary  |

FarmEx mainly focuses on customers who value:

- Freshness
- Food safety
- Chemical-free products
- Sustainable urban farming

#### Buyer Behavior

FarmEx customers are mostly urban, digitally active, and health conscious.

#### B2C Buyer Behavior

Customers usually:

- Search online before purchasing
- Follow health and organic food trends
- Prefer trusted and traceable food sources
- Are influenced by social media and food safety concerns
- Value freshness and direct delivery

#### B2B Buyer Behavior

Restaurants and retail businesses mainly prioritize:

- Consistent supply
- Product quality
- Freshness
- Reliable delivery
- Long-term supplier relationships

For B2B buyers, quality and trust are often more important than price.

## Competitor Analysis

| Competitor | Type            | Strength              | Weakness                     | FarmEx Advantage               |
|------------|-----------------|-----------------------|------------------------------|--------------------------------|
| Chaldal    | Online grocery  | Fast delivery         | No direct production control | FarmEx produces its own food   |
| Shwapno    | Supermarket     | Strong brand presence | Traditional supply chain     | Fresher rooftop-grown products |
| Agora      | Supermarket     | Large customer base   | Possible pesticide concerns  | Chemical-free guarantee        |
| Unimart    | Premium grocery | Premium positioning   | Higher prices                | More affordable fresh supply   |

FarmEx differentiates itself through:

- Smart aquaponics technology
- IoT monitoring
- Direct rooftop-to-door delivery
- Chemical-free production
- Dual revenue model

## Estimates of Annual Sales and Market Share

### Revenue Sources

Fish Production Revenue 600 kg yearly  $\approx$  ₳3,60,000

Vegetable Production Revenue  $\approx$  ₳40,000

Total Annual Revenue per Unit ₳4,00,000

FarmEx expects steady market growth as rooftop partnerships and customer subscriptions increase over time.



## 5. The Economics of the Business

### Revenue Drivers and Profit Margins

FarmEx generates revenue from:

- Fresh vegetable & fish sales
- Rooftop farm setup services
- Maintenance & training services

| Revenue Source         | Gross Margin |
|------------------------|--------------|
| B2C Food Sales         | ~35%         |
| B2B Supply             | ~34%         |
| Farm Setup Services    | ~46%         |
| Maintenance & Training | ~57%         |

Average rooftop unit profit increases after recovery of initial setup cost.

### Fixed and Variable Costs

#### Fixed Costs (Initial Setup Cost)

| Category                                 | Amount    |
|------------------------------------------|-----------|
| Rooftop Farming Setup                    | ₹4,00,000 |
| Seeds & Fish Cost (1 <sup>st</sup> year) | ₹1,80,000 |
| Miscellaneous & Installation             | ₹20,000   |

**Total Fixed Cost → ₹6,00,000 /Yearly**

#### Variable Costs

| Category             | Cost    |
|----------------------|---------|
| Seeds & Fish         | ₹60,000 |
| Water & Electricity  | ₹10,000 |
| Packaging & Delivery | ₹10,000 |

**Approx Variable Cost → ₹80,000**

### Operating Leverage and Its Implications

FarmEx has high initial setup costs but lower recurring operational costs.

As rooftop farm units increase:

- Profitability improves rapidly
- Fixed costs are shared across multiple farms
- IoT system becomes more cost-efficient
- Revenue scales faster than operating cost

## Start-up Costs

| Category               | Amount (BDT) |
|------------------------|--------------|
| Rooftop Farming Setup  | ₹4,00,000    |
| Seeds & Fish Cost      | ₹1,80,000    |
| Marketing & Operations | ₹20,000      |

**Total Start-up Cost → Approx. ₹6,00,000**

## Break-Even Calculation

**Break-Even Point = Total Initial Investment / Annual Net Profit**

- Initial Investment = ₹6,00,000
- Annual Revenue = ₹4,00,000
- Annual Operational Cost = ₹80,000

Annual Net Profit

Net Profit = Revenue – Operational Cost  
 = 4,00,000 – 80,000  
 = ₹3,20,000

Break-Even Calculation

Break-Even Point = 6,00,000 / 3,20,000  
 = 1.875 years  
 ≈ 2 years

From **Year 3 onward**, each rooftop farm unit generates stable recurring profit with relatively low operational cost.

## 6. Marketing Plan

### Overall Marketing Strategy

FarmEx's marketing strategy is built on three core pillars:

- **Trust & Transparency:** Live farm monitoring, traceability, and customer trust building through real farm data
- **Community & Education:** Creating awareness through sustainable farming content, videos, and educational campaigns
- **Digital-First Growth:** Strong focus on Facebook, Instagram, WhatsApp, and referral-based word-of-mouth marketing

FarmEx positions itself as a “সবুজ যখন ঘরের অতিথি” in Dhaka's urban food market.

### Product, Price, Promotion, and Place (4Ps)

#### Product

- Fresh vegetables and fish from rooftop aquaponics farms
- Eco-friendly packaging (jute bags / biodegradable materials)
- QR code system for farm traceability (harvest date + farm info)
- Brand promise: “**Rooftop to Table in 45 min**”

#### Price

| Product / Service               | Price (BDT)  | Positioning            |
|---------------------------------|--------------|------------------------|
| Weekly Veg Box (5-8 kg)         | 600          | Value premium          |
| Weekly Family Box (veg + fish)  | 1,200        | Premium                |
| Monthly Veg Subscription        | 2,200        | Loyalty plan           |
| Monthly Veg + Fish Subscription | 4,200        | Premium plan           |
| Farm Fish (per kg)              | 200          | Above market (280–320) |
| Rooftop Setup Service           | 4,00,000     | One-time service       |
| Maintenance Contract            | 10,000/month | Recurring service      |

#### Promotion

- Farm visit campaign
- Referral program (free box for referrals)
- Influencer marketing (food, health, parenting niches)
- Educational content on Facebook & YouTube
- Restaurant branding: “Grown on FarmX Rooftop”

#### Place (Distribution)

- Direct home delivery inside Dhaka
- FarmX mobile app and website ordering system
- Restaurant and café supply chain (B2B weekly delivery)

## **Sales Process (Cycle)**

1. Customer discovers FarmEx via social media or referral
2. Visits website/app and explores products
3. Converts to weekly/monthly subscription
4. Engages with FarmEx content and updates
5. Refers others using referral program

## **Sales Tactics**

- Free/discounted first trial box for new customers
- Loyalty points system for repeat customers
- Corporate health packages for companies
- WhatsApp community groups for updates and engagement
- Subscription-based predictable income model

## 7. Design and Development Plan

### Development Status and Tasks

| Phase   | Timeline   | Key Tasks                                                                                 | Status  |
|---------|------------|-------------------------------------------------------------------------------------------|---------|
| Phase 1 | Month 0–3  | Pilot farm setup, aquaponics system testing, IoT sensor installation, core team formation | Planned |
| Phase 2 | Month 4–6  | Mobile app development, 1 additional rooftop farms, delivery system setup, brand launch   | Planned |
| Phase 3 | Month 7–12 | Commercial launch, 120 subscribers target, 2 B2B contracts, 2 setup projects              | Planned |
| Phase 4 | Year 2     | Scale to 5 farms, AI-based app upgrade, expansion to Dhaka zones                          | Planned |

### Challenges and Risks

| Risk                     | Description                                   | Mitigation                                                 |
|--------------------------|-----------------------------------------------|------------------------------------------------------------|
| Weather Risk             | Monsoon and seasonal damage to rooftop farms  | Weather-resistant structures and controlled farming system |
| IoT Failure              | Sensor malfunction may affect monitoring      | Backup sensors and manual checking system                  |
| Rooftop Access Risk      | Owners may withdraw rooftop space             | Long-term contracts with revenue sharing model             |
| Market Price Sensitivity | Customers may hesitate due to premium pricing | Focus on health value + subscription discounts             |
| Delivery Risk            | Late delivery affects customer trust          | Own delivery team with scheduled zones                     |

### Projected Development Costs

| Development Item                          | Cost (BDT)    |
|-------------------------------------------|---------------|
| Mobile App (iOS + Android)                | We will do it |
| Website & E-commerce Platform             | We will do it |
| IoT Dashboard & Backend                   | We will do it |
| Rooftop Farm Setup + Aquaponics (1 units) | ₳ 5,00,000    |

**Total Development Investment → ₳ 5,00,000**

### Proprietary Issues

- **Brand Name:** “FarmEX” registered with DPDT Bangladesh
- **Software Ownership:** Mobile app, IoT system
- **Trade Secrets:** Aquaponics system design, crop-fish ratio, and IoT integration methods
- **Digital Assets:** Website and social media handles secured (farmexbd.com, @FarmEXBD)
- **Intellectual Property:** All system designs and operational models are proprietary to FarmEX Pvt. Ltd.

## 8. Operations Plan

### General Approach to Operations

FarmEx operates as a **vertically integrated urban farming system**, managing three core areas:

- **Farm Production:** Rooftop-based aquaponics farming for fish and vegetables
- **Technology Management:** IoT-based real-time monitoring system for water quality and farm health
- **Last-Mile Delivery:** Direct delivery of fresh produce to customers within Dhaka

Each rooftop farm functions as an independent production unit, managed by a site technician responsible for:

- Daily farm monitoring
- Fish feeding and plant care
- Basic maintenance and reporting

### Facilities and Equipment

#### Per Rooftop Farm Site

| Item         | Specification                                   |
|--------------|-------------------------------------------------|
| Fish Tanks   | 10,000L tanks (Tilapia / Koi fish)              |
| Water System | Submersible pump + biofilter system             |
| IoT Sensors  | pH, temperature, oxygen, water level monitoring |
| Structure    | Shade net + weather-resistant rooftop framing   |
| Storage      | Harvest crates + insulated cooling boxes        |

## 9. Management Team and Company Structure

### Management Team

FarmEx is built and managed by a core founding team of four members, where each member plays an equal role in the company. The ownership of the company is equally distributed among all founders.

#### Equity Structure

- Each founder holds **25% ownership share** of FarmEx

### Core Management Team

| Name               | Role                           | Responsibility                                                                              |
|--------------------|--------------------------------|---------------------------------------------------------------------------------------------|
| Kamrul Hasan Siam  | CEO / Co-Founder               | Leads overall strategy, business planning, and smart rooftop farming integration.           |
| Momin Ali Rifat    | CTO / Co-Founder               | Manages IoT systems, automation, and technical infrastructure                               |
| Nazmuz Saif        | Operations Manager/ Co-Founder | Handles rooftop farming operations, logistics, supplier coordination, and product delivery. |
| Tahsin Rahman Khan | Marketing Manager/ Co-Founder  | Responsible for branding, customer acquisition, partnerships, and market expansion.         |

### Skills Coverage Analysis

| Skill Area                   | Responsible Person                     | Notes                             |
|------------------------------|----------------------------------------|-----------------------------------|
| Business Strategy & Finance  | CEO (Kamrul Hasan Siam)                | Overall leadership and planning   |
| Technology & IoT Development | CTO (Momin Ali Rifat)                  | Full technical system development |
| Farm Operations & Logistics  | Operations Manager (Nazmuz Saif)       | Day-to-day operations management  |
| Marketing & Customer Growth  | Marketing Manager (Tahsin Rahman Khan) | Digital marketing and branding    |

## 10. Overall Schedule

| Milestone                      | Target                |
|--------------------------------|-----------------------|
| Company Registration (RJSC)    | Planned (Month 1)     |
| Seed Investment                | Planned (Month 1–2)   |
| (App + IoT System) Development | Planned (Month 4)     |
| 2 Additional Farm Sites Setup  | Planned (Month 5)     |
| Brand & Marketing Launch       | Planned (Month 5)     |
| First Commercial Deliveries    | Planned (Month 6)     |
| 50 Active B2C Subscribers      | Planned (Month 7)     |
| First B2B Supply Agreement     | Planned (Month 7)     |
| First Rooftop Setup Project    | Planned (Month 8)     |
| 120 Active B2C Subscribers     | Planned (Month 12)    |
| Operating Break-Even           | Planned (Month 16–18) |
| 5 Farm Sites Operational       | Planned (Month 20)    |
| 380 Active Subscribers         | Planned (Month 24)    |
| Full Profitability             | Planned (Month 28–30) |



## 11. Financial Projections

**Total Seed Funding Required: BDT 7,00,000**

### Sources

| Source            | Amount (BDT)    | Type   |
|-------------------|-----------------|--------|
| Founders' Capital | 1,00,000        | Equity |
| Friends & Family  | 2,00,000        | Equity |
| <b>Total</b>      | <b>3,00,000</b> |        |

### Uses

| Use                          | Amount (BDT)    |
|------------------------------|-----------------|
| Smart Rooftop Farming Setup  | 5,00,000        |
| Initial Seeds & Fish Inputs  | 60,000          |
| Buffer (Ops + setup support) | 40,000          |
| <b>Total</b>                 | <b>6,00,000</b> |

### Assumptions Sheet (Per Farm Unit)

- Rooftop model with **profit-sharing (15%–20%)**
- Initial setup cost: **₹6,00,000**
- Fish revenue: **600 kg/year ≈ ₹3,60,000**
- Vegetable revenue: **₹40,000/year**
- Total annual revenue: **₹4,00,000**
- Annual operational cost: **₹80,000**
- Annual profit per unit: **₹3,20,000**
- Monthly income per unit: **~₹30,000**
- Tech service margin: **15% on setup sales**

### Ratio Analysis

#### 1. Profit per Unit

**Annual Revenue – Annual Cost**

$$= 4,00,000 - 80,000$$

$$= 3,20,000$$

#### 2. ROI (Return on Investment)

**(Annual Profit ÷ Initial Investment) × 100**

$$= (3,20,000 \div 6,00,000) \times 100$$

$$= 53.33\% \approx 53\%$$

#### 3. Break-Even Time

**Initial Investment ÷ Annual Profit**

$$= 6,00,000 \div 3,20,000$$

$$= 1.875 \text{ years } (\sim 2 \text{ years})$$

#### 4. Net Margin

$$(\text{Profit} \div \text{Revenue}) \times 100$$

$$= (3,20,000 \div 4,00,000) \times 100$$

$$= 80\%$$

#### Key Insight

- FarmEX is a **unit-based scalable model**
- Each rooftop generates **₹3,20,000 annual profit**
- Break-even achieved in **~2 years**
- Growth depends on **adding more rooftop units, not increasing fixed cost heavily**
- Tech + setup service creates **additional revenue layer (15% margin)**

## 12. Value Chain

### Core Activities

| Stage                        | Activities                                                                                                                                                                         |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inbound Logistics</b>     | Rooftop space partnerships, organic seeds procurement, fish fry supply, nutrients & bio-fertilizers sourcing, IoT sensors & smart farming equipment procurement                    |
| <b>Operations</b>            | Rooftop aquaponics farm setup, IoT-based monitoring system, automated water circulation system, integrated fish & vegetable cultivation, growth tracking & data analytics          |
| <b>Outbound Logistics</b>    | Fresh packaging of vegetables & fish, home delivery (B2C), supply to restaurants & retailers (B2B), fast delivery-focused minimal cold-chain logistics                             |
| <b>Marketing &amp; Sales</b> | Social media marketing (Facebook, Instagram, TikTok), SEO & website awareness, referral & community campaigns, restaurant partnerships, educational content on sustainable farming |
| <b>After-Sales Services</b>  | Customer feedback & review system, subscription management, complaint handling & replacement support, loyalty rewards, rooftop setup client support                                |

### Support Activities

| Area                             | Activities                                                                                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Infrastructure</b>            | Legal & compliance (food safety, urban farming), business administration & finance management                           |
| <b>Human Resource Management</b> | Agronomists, IoT engineers, software developers, farm technicians & maintenance staff                                   |
| <b>Technology Development</b>    | IoT smart farming system, mobile app for monitoring & subscription, data analytics for yield optimization               |
| <b>Resource Procurement</b>      | Organic seeds & fish fry suppliers, IoT hardware vendors, farming equipment suppliers, eco-friendly packaging materials |

## 13. Business Model

### Business Model Canvas - FarmX

